Students' activity

Place the appropriate number given in the diagram in the bracket

Lacolith ( )

Batholith ( )

Dyke ( )

Sill ( )

Chamber ( )

Lapolith ( )

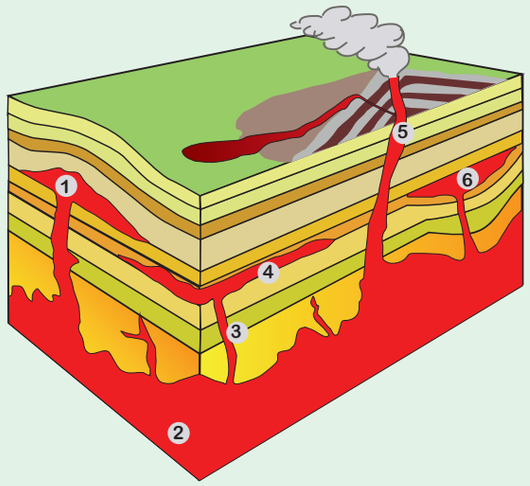


Figure 3.27 Intrusive Volcanic Structures

Table 3.2 Metamorphosis processes

Igneous/ Sedimentary rock	Influence	Metamorphosed rock
Granite	Pressure	Gneiss
Clay, Shale	Pressure	Schist
Sandstone	Heat	Quartzite
Clay, Shale	Heat	Slate or Phyllite
Coal	Heat	Anthracite or Graphite
Coal	Heat and Pressure	Diamond
Limestone	Heat	Marble

Metamorphism is a process by which the already consolidated rocks undergo recrystallisation and reorganization of materials within original rocks. Gneiss, slate, schist, diamond, marble, quartzite etc. are some examples of metamorphic rocks. The igneous and metamorphic rocks together account for 95 percent of the earth.

3.12 Rock Cycle

Rock cycle is a continuous process through which old rocks are transformed into new ones. Igneous rocks can be changed into sedimentary or metamorphic rocks. The fragments derived out of igneous and metamorphic rocks form into sedimentary rocks.

Igneous and sedimentary rocks can change into metamorphic rocks. The crustal rocks (igneous, sedimentary and metamorphic) may be carried down into the mantle (interior of the earth) through subduction process and the same melt and turn into magma, the original source for igneous rocks. In this way the rock cycle is a continuous process.

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